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unsupervised learning - part 2

dimensionality reduction

- feature elimination
- feature extraction

principal component analysis

- feature extraction
- produces low-dimensional representation of dataset
- seq. of linear combinations of variables with maximal variance and mutually uncorrelated
- assist in data visualisation

Consider the features $X_1, X_2, \dots, X_p$

$$Z_1 = \theta_{11}X_1 + \theta_{21}X_2 + \dots + \theta_{p1}X_p$$

$$\sum_{j=1}^p \theta_{j1}^2 = 1$$

$\theta_{11}, \theta_{21}, \dots, \theta_{p1}$ **are the loadings of the principal component**

$$\theta_1 = [\theta_{11} \ \theta_{21} \ \dots \ \theta_{p1}]^T$$

$$\mathbf{size}(X) = n \times p$$

$$\mathbf{maximize}_{\theta_{11}, \theta_{21}, \dots, \theta_{p1}}$$

$$\mathbf{subject\ to}$$

$$\frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \theta_{j1} x_{ij} \right)^2$$

$$\sum_{j=1}^p \theta_{j1}^2 = 1$$

procedure

- standardize variables
- compute co-variance matrix
- find eigenvectors and corresponding eigen values of co-variance matrix

$$\text{cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)(y_i - \mu_y)$$

$$A \cdot V = \lambda \cdot V$$

$$\det(A - \lambda I) = 0$$

note...

- the loading vector defines the direction along which the data varies the most
- projecting n data points in that direction shows the principal component scores
- the second principal component follows the same procedure but is uncorrelated (orthogonal) with the first one.

- strength of each component is determined by the **proportion of variance explained** (PVE)

$$\sum_{j=1}^p Var(X_j) = \sum_{j=1}^p \frac{1}{n} \sum_{i=1}^n x_{ij}^2$$

$$Var(Z_\ell) = \frac{1}{n} \sum_{i=1}^n Z_\ell^2$$

$$\frac{\sum_{i=1}^n Z_{i\ell}^2}{\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2}$$

Gaussian mixture models

- assumes data represented by multiple normal distributions
 - mean (centre)
 - co-variance (width)
 - mixing probability (strength)

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{D}{2}} \Sigma^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$$

$$p(x_n) = \sum_{k=1}^K \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k)$$

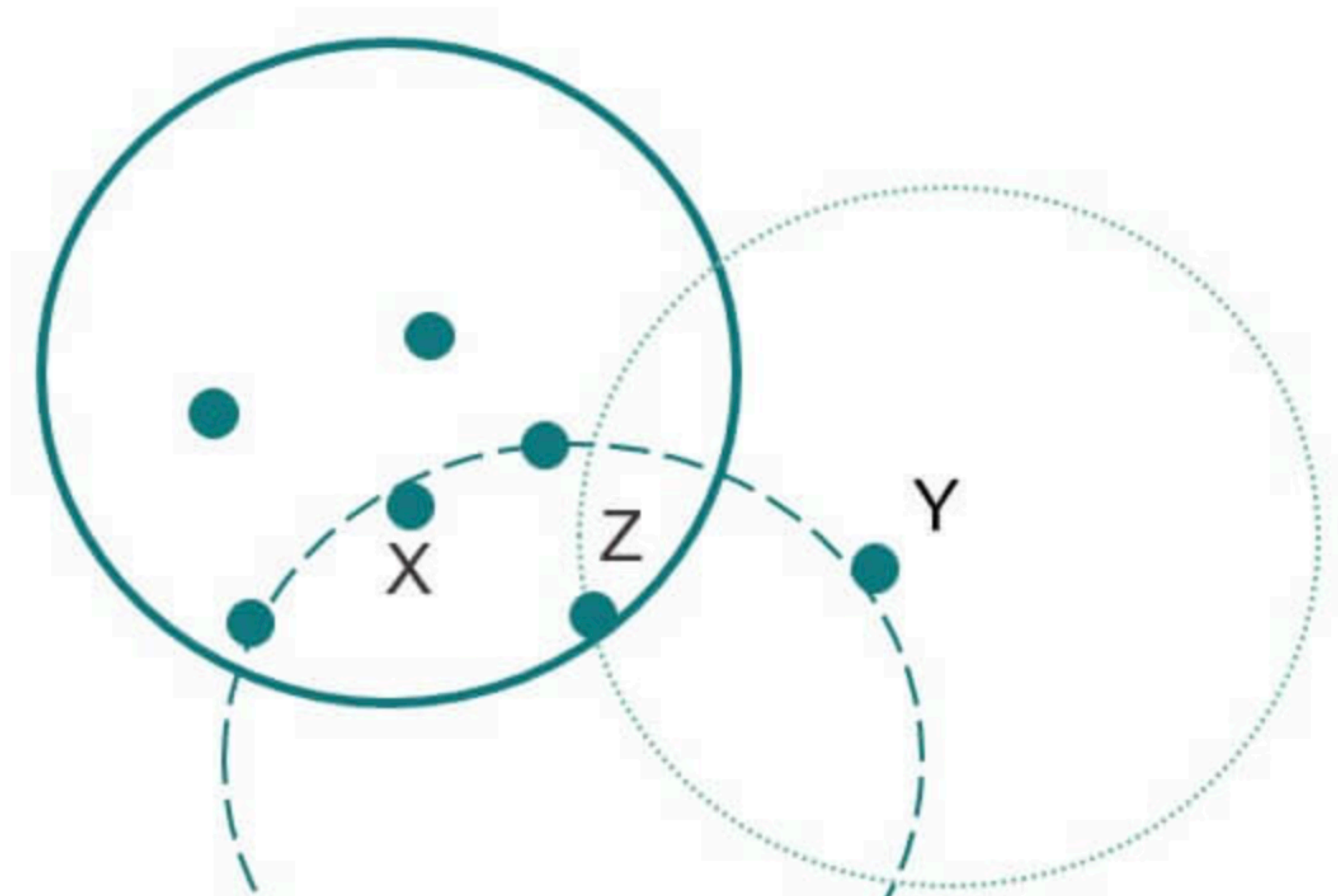
- maximum likelihood
- expectation-maximisation algorithm to determine data point probability distribution in a cluster

- Illustration:
 - see notebook

density-based algorithms

- discover clusters of arbitrary shapes
- distinguish noise

- group data points based on their density
- core points
 - minimum number of neighbors in a radius
- connect core points to neighboring points



in short...

- clustering
 - k-means
 - hierarchical
 - model-based (gaussian mixture)
 - density-based
- dimensionality reduction
 - principal component analysis